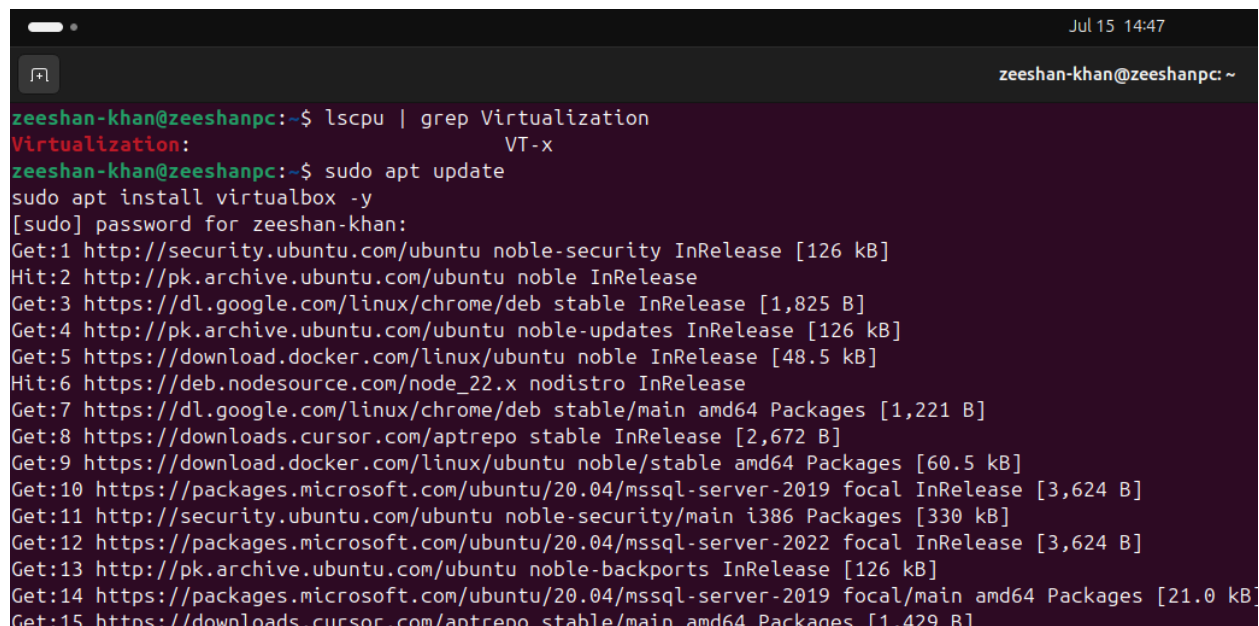


1. Introduction

Objective:

The objective of this task was to install Ubuntu Server (CLI version) on a virtual machine using VirtualBox. This activity helped in understanding virtualization and Linux system administration.

Step 1: Downloading VirtualBox

A terminal window screenshot with a dark background. The title bar shows 'Jul 15 14:47' and the user 'zeeshan-khan@zeeshanpc: ~'. The terminal shows the following commands and output:

```
zeeshan-khan@zeeshanpc:~$ lscpu | grep Virtualization
Virtualization:          VT-x
zeeshan-khan@zeeshanpc:~$ sudo apt update
[sudo] password for zeeshan-khan:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 https://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:6 https://deb.nodesource.com/node_22.x nodistro InRelease
Get:7 https://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,221 B]
Get:8 https://downloads.cursor.com/aptrepo stable InRelease [2,672 B]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [60.5 kB]
Get:10 https://packages.microsoft.com/ubuntu/20.04/mssql-server-2019 focal InRelease [3,624 B]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [330 kB]
Get:12 https://packages.microsoft.com/ubuntu/20.04/mssql-server-2022 focal InRelease [3,624 B]
Get:13 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:14 https://packages.microsoft.com/ubuntu/20.04/mssql-server-2019 focal/main amd64 Packages [21.0 kB]
Get:15 https://downloads.cursor.com/aptrepo stable/main amd64 Packages [1,429 B]
```

Screenshot:

Figure 1: VirtualBox official download page

Step 2: VirtualBox main interface after successful installation

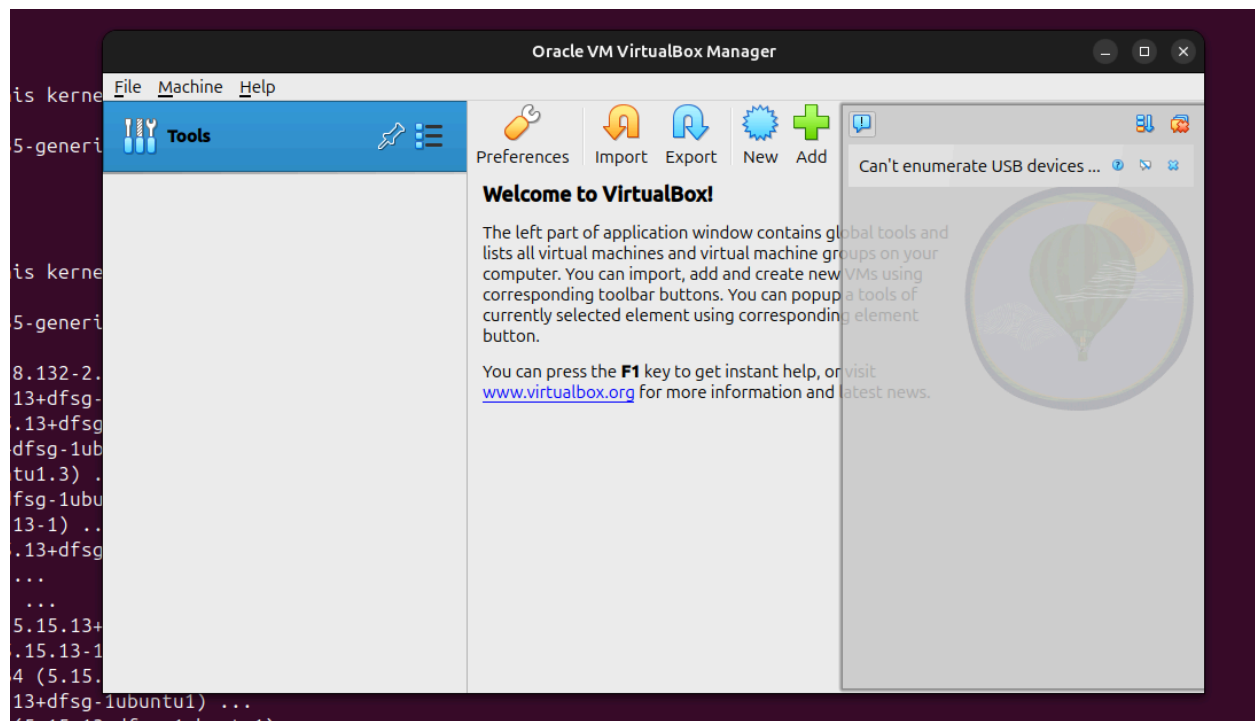
Oracle VirtualBox was installed on my Ubuntu system using the terminal and Linux package management commands. The required VirtualBox packages were downloaded and configured directly through the command line environment.

The installation was performed using the following commands:

```
sudo apt update
```

```
sudo apt install virtualbox
```

After successful installation, the VirtualBox application was launched and verified.



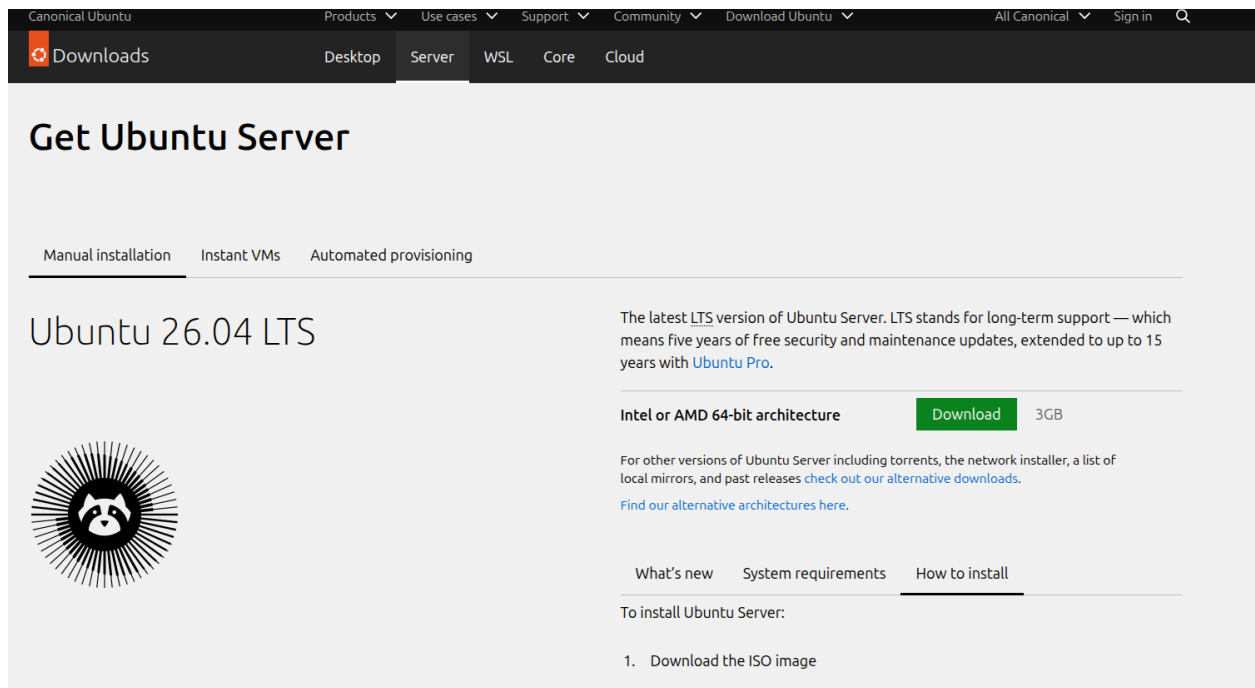
Screenshot:

Figure 1: Terminal showing VirtualBox installation process

Step 3: Downloading Ubuntu Server ISO

The Ubuntu Server ISO image was downloaded from the official Ubuntu website.

The latest Ubuntu Server LTS 64-bit version was selected because LTS releases provide:



Screenshots:

Figure 3: Ubuntu Server download page

Step 4: Creating a New Virtual Machine in VirtualBox

A new virtual machine named Ubuntu-Server was created in VirtualBox with the following configuration:

Configuration	Details
Virtual Machine Name	Ubuntu-server
Operating System	Linux

Version	Ubuntu (64-bit)
RAM Allocation	2048 MB (2 GB)
Storage	20 GB Dynamic Disk

Create Virtual Machine

Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Name:

Server

✓

Folder:

/home/zeeshan-khan/VirtualBox VMs

▼

ISO Image:

/home/zeeshan-khan/Downloads/ubuntu-24.04.4-live-server-amd64.iso

▼

Edition:

▼

Type:

Linux

▼

64

Version:

Ubuntu (64-bit)

▼

☐ Skip Unattended Installation

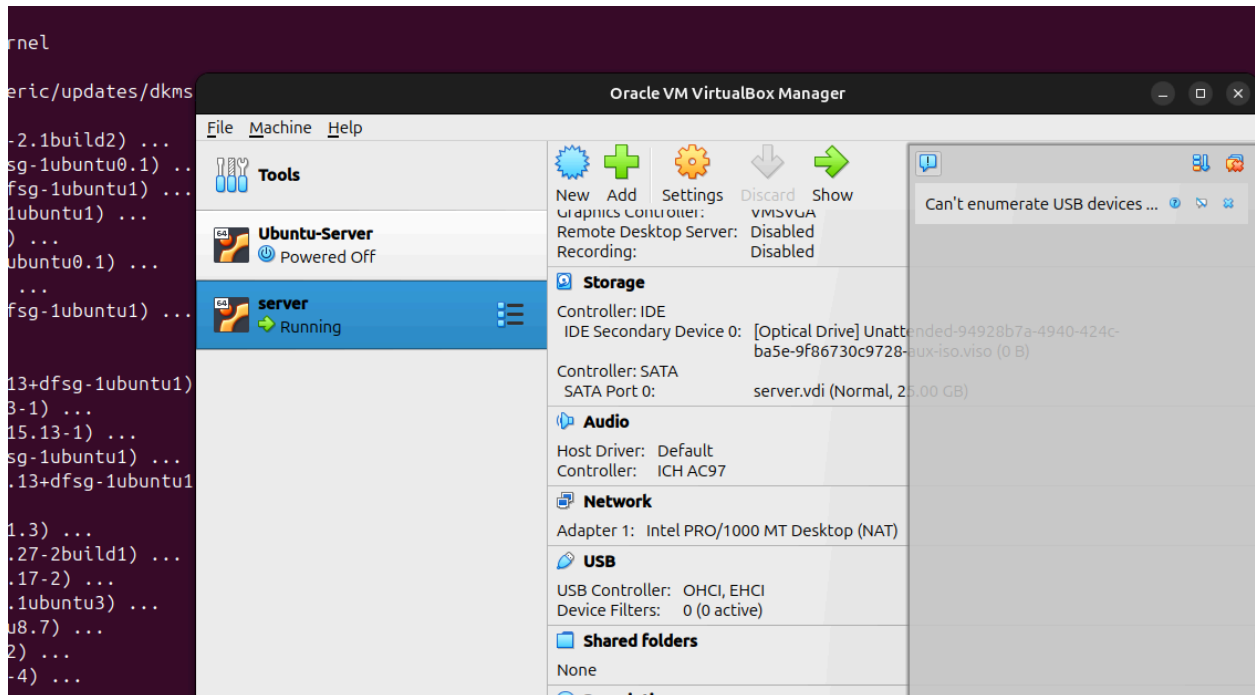
Detected OS type: Ubuntu (64-bit). This OS type can be installed unattended. The install will start after this wizard is closed.

Expert Mode

Back

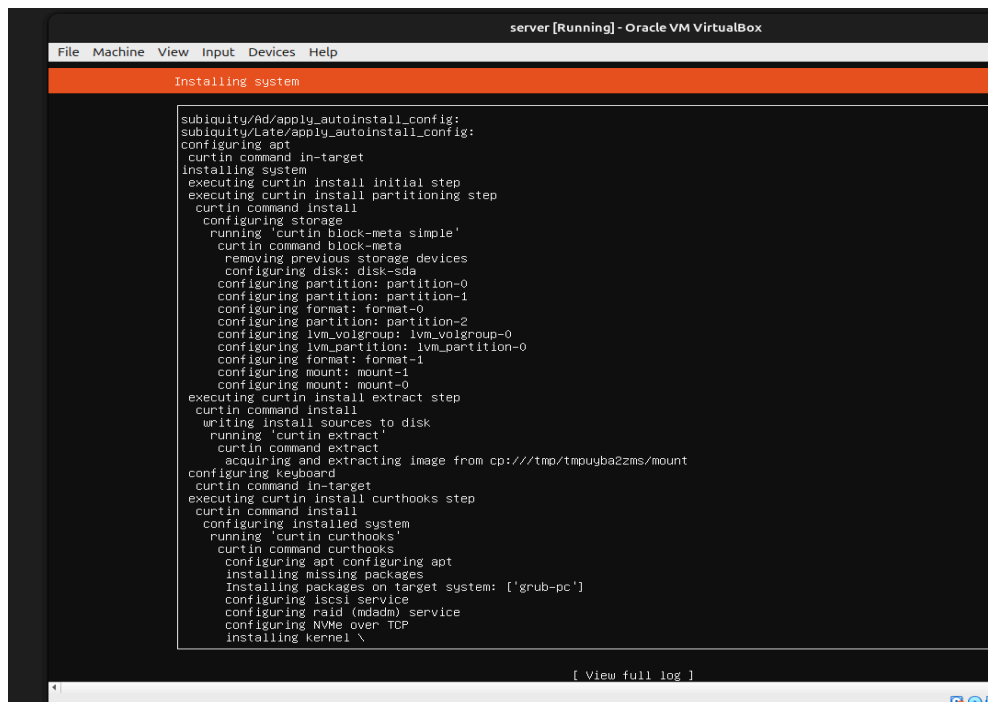
Next

Cancel



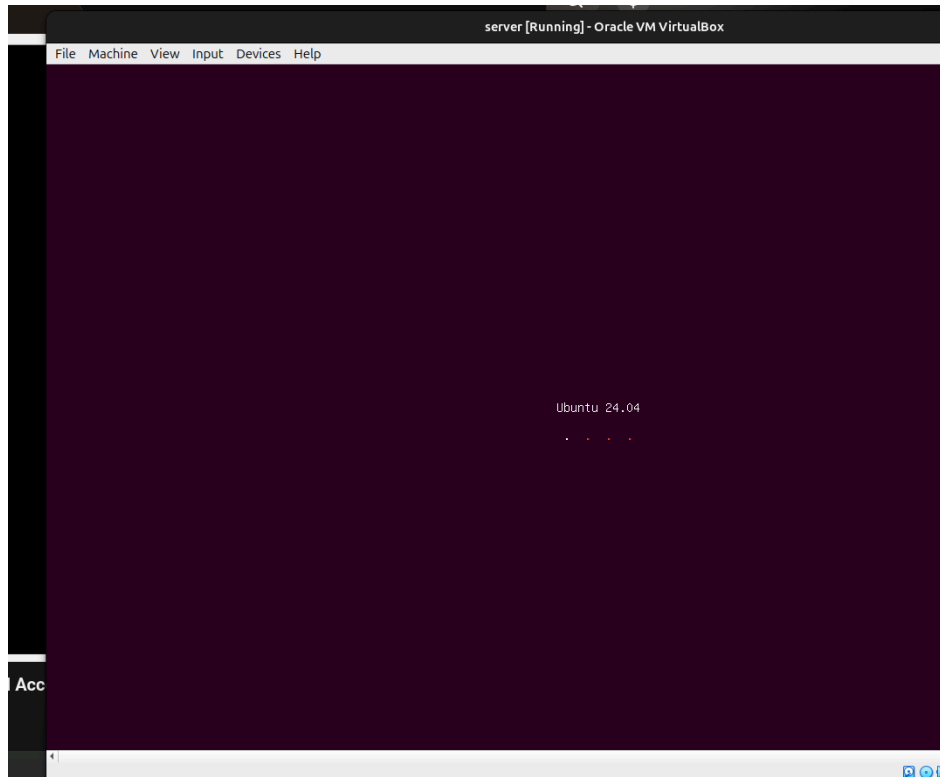
Step 5: Ubuntu Server Installation Process

The Ubuntu Server ISO was mounted to the virtual machine, and the installation process was started.



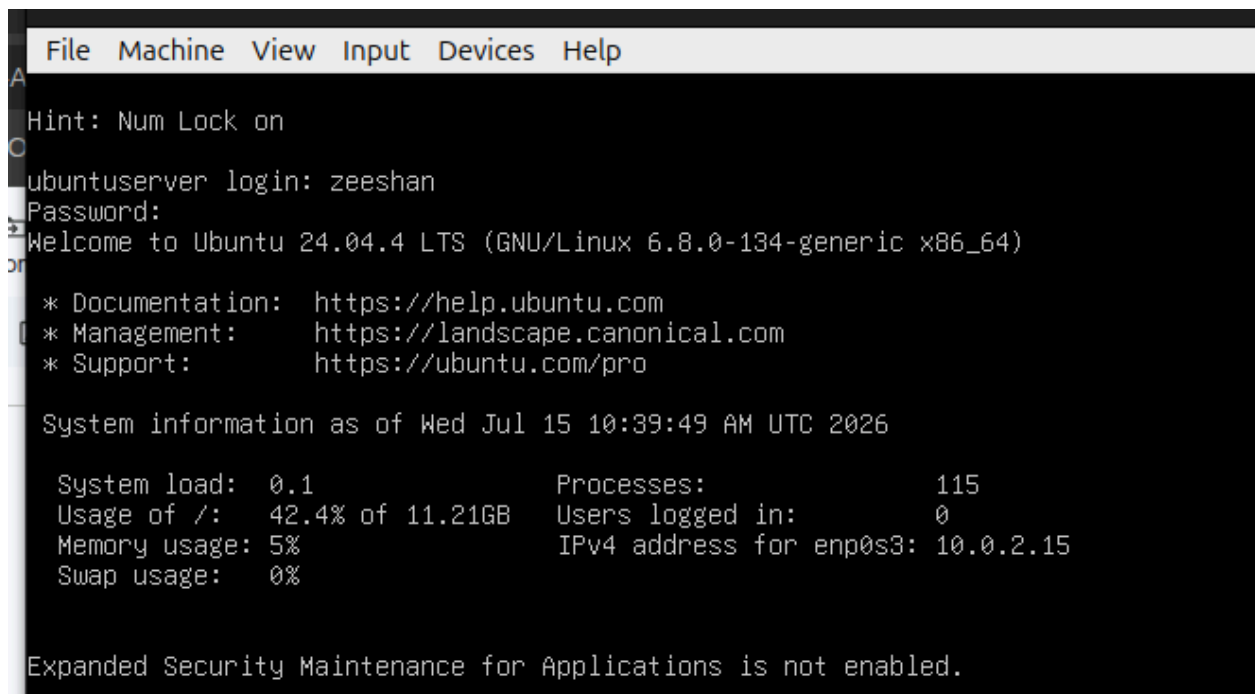
Step 6: System Reboot

After installation completion, the system requested a reboot.



Step 7: Successful Login

After rebooting, I logged into Ubuntu Server using the created user account



```

individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

zeeshan@ubuntuuser:~$ free -m
               total        used        free      shared  buff/cache   available
Mem:           4098         395        3629          1         292        3703
Swap:          2264           0        2264

zeeshan@ubuntuuser:~$
zeeshan@ubuntuuser:~$ free -m
               total        used        free      shared  buff/cache   available
Mem:           4098         395        3629          1         292        3703
Swap:          2264           0        2264

zeeshan@ubuntuuser:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
zeeshan@ubuntuuser:~$

```

```

server [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Ubuntu 24.04.4 LTS ubuntuuser tty1

Hint: Num Lock on

ubuntuuser login: zeeshan
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 15 10:39:49 AM UTC 2026

System load:  0.1               Processes:    115
Usage of /:   42.4% of 11.21GB  Users logged in: 0
Memory usage: 5%               IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

zeeshan@ubuntuuser:~$

```